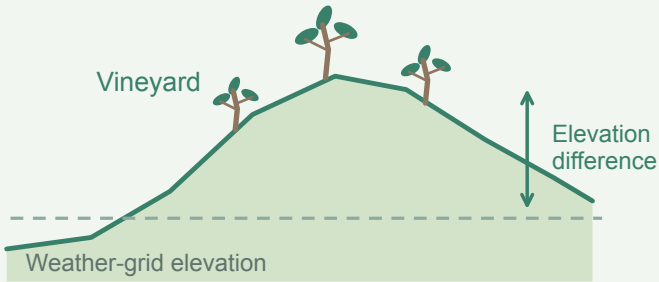
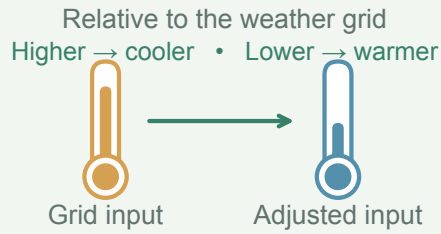
Grapevine phenology prediction

Five environmental zones (EZs) • Four developmental stages

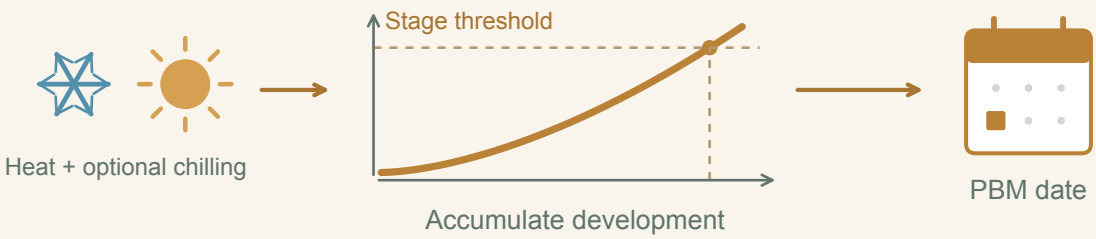
DEM Adjust temperature to vineyard elevation



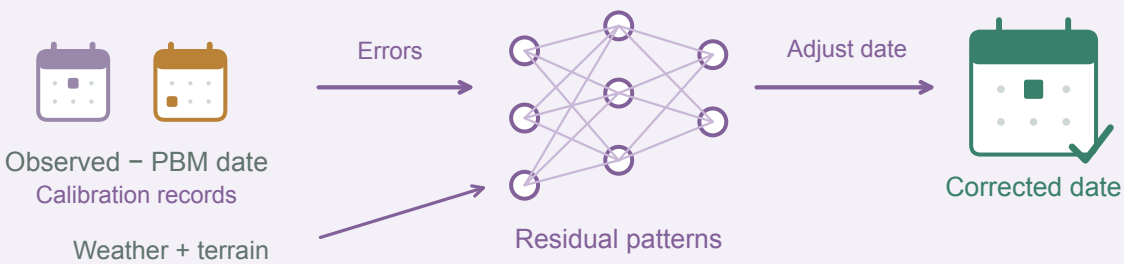


Reduce elevation-related mismatch in temperature forcing

PBMs Translate temperature into development



ML Learn and correct remaining date errors





DEM helps PBMs; little extra gain after ML.

Flowering ML gain: EZ05 10.58 d • EZ03 1.44 d

Gains depend on zone and stage; sparse coverage limits inference.

PBMs: process-based models • DEM: digital elevation model • ML: machine learning